INCH-POUND

MIL-R-50651B (AR)
26 December 1990
SUPERSEDING
MIL-R-50651A
30 August 1985

MILITARY SPECIFICATION

RADAR SET AN/VPS-2: 10549058
(RANGE ONLY RADAR)

This specification Is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for Use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers Radar Set AN/VPS-2, hereafter re erred to as the radar. The radar is used in the fire control system (FCS) for the 20-millimeter Antiaircraft (20-Jmn AA) Artillery Gun, M163A1 (self-propelled) and M167A1 (towed), associated with the forward Area Air defense System (FAADS) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the currenversion before use.

'MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-D-48435 - Distribution Box, 10548533
MIL-R-50668 - Receiver-Transmitter, Radar RT-860 10548089 (Unit 2)
MIL-R-50674 - Receiver, Radar R-1475 10548338 (Unit 3)
MIL-C-50685 - Computer, Range CP-888 10548339 (Unit 4)
MIL-P-50686 - Power Supply, PP-4812 10548440 (Unit 5)
MIL-A-50687 - Antenna Assembly, AS-2049 10548388 (Unit 1)

STANDARDS

MILITARY

- MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-81O - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-50940)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548164 IF Amplifier
10549058 Radar Set AN/VPS-2
11748124 System Tester, Radar AN/VPM-2

PACKAGING DATA SHEET

PS10549O58 - packaging of Radar Set, AN/VPS-2

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the currentversion before use.

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall **comply** with all requirements specified on Drawing 10549058, all associated drawings, and with all requirements specified in applicable specifications and standards. The units of the radar shall have been inspected and accepted in accordance with their respective drawings or specifications prior to their installation in the radar. The following units are covered by the indicated **specifications:**

UNIT NO.	NAME	SPECIFICATION
1	**Antenna Assembly AS-2049, 10548388**	**MIL-A-50687**
2	**Receiver-Transmitter, Radar RT-860, 10548C)e9**	**MIL-R--50668**
3	**ReceiVer, Radar R 1475, 10548998**	**MIL-R-50674**
4	Computer, Range CP-888, 10548339	**MIL-C-50685**
5	Power Supply PP-4812, 10548440	**MIL-P-50686**
6	**Distribution Box, 10548533**	**MIL-D-48435**

Where applicable, the radar shall meet the requirements of MIL-F-13926 as **follows:**

a. **Dimensions and tolerances**
b. **Inorganic protective surface finishings**
c. **part identification and marking**
d. **Workmanship**

Source: https://assist.dla.mil -- Downloaded: 2026t0-22T20:31Z
Check the source to verify that this is the currentversion before use.

3.3 Environmental service conditions.

3.3.1 vibration. The radar shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.-3.2 Shock. The radar shall not suffer damage and shall meet the requirments of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ~+10%, 11 milliseconds ~10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude,

The radar shall meet the requirements of 3.4 and 3.5 during an after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-Operating. The radar shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The radar shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg.F and +155 deg.F.

3.3.4.2 Non-operating (storage) . The radar shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg.F for a period of 24 hours and +160 deg.F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The radar shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the radar.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

3.5 Performance. The radar shall meet the requirements of this section when supplied with 22 volts, direct current (Vdc) to 30 Vdc from a 750-watt (W) source and 115 V ± 5%, 400 Hz 11%, from a **15 W source.**

3.5.1 Standby power. The radar shall use the 22 Vdc to 30 Vdc primary power to activate its standby circuits. The blowers in units 2 and 5 shall be activated upon application of standby power . A **2-minute time delay relay shall be activated by application** of standby power which shall enable the radiate function 120 +30, -15 second(s) after applicatioon of standby power.

3.5.2 Radar ready signal. The radar shall provide a single pole double-throw (SPDT) switch action to the FCS, indicating that the radar is ready-to radiate.

3.5.3 Transmit rf indication. The radar shall have a visual indicator to show that (radio frequency) rf energy is being transmitted from the rf transmitting system. The transmit indicator shall be generated by the receiver-transmitter (RT) unit and the transmit indicator shall be mounted so that it is visible to the gunner.

3.5.4 Radiate function. The radiate function shall be available through closure of 2S6 (SERV RAD CONT) in the RT unit **cover or through application of +22 Vdc to +30 vdc at 231 pin R from a remote control unit.**

3.5.5 Unit 2 interlock. The unit 2 interlock switch shall close when the cover of the RT unit is tightened down enabling unit 2 operation. When the RT unit cover is open, the interlock switch shall open, preventing operation of unit 2. **The manual override feature shall permit operation of unit 2 when the RT unit cover is open and the interlock switch is in the manual override position.**

3.5.6 Transmit characteristics.

3.5.6.1 Average Power. The minimum average transmit power on any channel shall be 4.26 W at 2J3.

3.5.6.2 Rf transmit pulse.

a. **The radar transmit pulse shape shall be cosine squared (COS^2).**
b. The radar transmit pulse width shall be 280~20 nano seconds (ns) at 50 percent points

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

3.5.6.3 Transmit frequency. The receiver-transmitter unit shall be capable of operation on six discrete frequencies by selecting one of six oscillator plug-in modules. The frequency of the transmitted signal shall be 9205 megahertz (MHz) ± 1 0 MHz using charnel 1 oscillator module. Using the channel I measurement as a reference, the channel 2 through 6 output frequency shall be as follows:

Chan. 2	Reference + 8 52 MHz
Chan. 3	Reference + 162 MHz
Chan. 4	Reference + 24 ~2 MHz
Chan. 5	Reference + 32 52 MHz
Chan. 6	Reference + 40 2 MHz

3.5.6.4 PUlse repetition frequency. The PRF shall be not less than 19,900 and not more than 20,100 pulses per second.

3.5.7 Mid-range cal function.. The mid-range cal function shall -provide a self test of the radar ranging circuits. When mid-range cal switch (4S1) on the range computer is depressed, mid-range cal lamp (4DS1) shall illuminate brightly.

3.5.8 System noise figure. The system noise figure shall be 13 dB maximum at room temperature (70 deg.F).

3.5.9, Receiving .charactaristics.. .

3.5.9.1 Clutter lock-on function. The clutter lock-on unction shall enable radar lock onto stationary targets for test urposes when the clutter lock-on switch on unit 2 is in the test osition.

3.5.9.2 Range reference voltage. The radar shall supply a +20 ±200 mV reference signal at 4J2-H(+) and 4J2-K(-) to the FCS.

3.5.9.3 Range standby voltage. When the range computer is in he search mode, a standby range analog voltage of 6.4 V ±0.1 V hall be supplied at 4J2-J(+) and 4J2-K(-).

3.5.9.4 Range rate positive reference voltage. The radar hall provide a positive range rate reference voltage of 10 V 200 mV at 4J2-B(+) and 4J2-L(-).

3.5.9.5 Range rate negative reference voltage. The radar hall provide a negative range rate reference voltage of -10 V 200 mV at 4J2-C(-) and 4J2-L(+).

3.5.9.6 Range rate standby voltage. When the range computer s in the search mode, a standby range rate voltage of -6.4 V ±.01 shall be supplied at 4J2-A(-) and 4J2-L(+).

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the currenversion before use.

3.5.9.7 Range accuracy. The accuracy of the range voltage supplied at 4J2-J(+) and 4J2-K(-) shall be 4V/1000 meters ~40 mV for targets of 1000 meters range or less and 4V/1000 meters ~1% for targets at greater than 1000 meters to the maximum range of 5000 meters.

3.5.9.8 Subclutter visibility. The subclutter visibility shall be a minimum of 40 dB with or without AGC (Automatic Gain Control) under MDS (Minimum Detectable Signal) conditions. The subclutter visibility with AGC at a clutter level of -15 dBm shall be 40 dB minimum.

3.5.9.9 Agc characteristics. With a clutter signal input, the AGC cut-in Doint shall be 38 to 40 dB above the MDS level. For 10 dB increments in clutter power from the AGC cut-in point, AGC action shall reduce the gain of the receiver by 10 dB per increment with the allowable tolerance given i.n procedure 4.6.3.9.11.

3.5.9.10 Active ECM indication. The radar shall provide a contact closure to activate the remote panel ECM lamp when the signal level is 20/5 dB greater than the lock threshold and the target velocity is less than 10 M/s.

3.5.9.11 Range rate accuracy. The range rate voltage between 4J2-A and 4J2-L(RET) shall meet the requirements shown in Table I for target velocities show in Table I.

TABLE I. R81XW rate accuracy requIrementm

VELOCITY ~	RANGI? RATE REF. [V)	kATIO tiT	NMT
Oumotnm ' ' ' '			
00918.7	+10	0.888	0.921
00697.2	+10	0.660	0.693
00344.2	+10	0.323	0.355
00057.4	+10	0.0404	0.0727
INBOUND			
00057.4	-lo	0.0404	0.0727
00344.2	-lo	0.323	0.355
00697.2	-lo	0.660	0.693
00918.7	-lo	0.888	0.921

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the currenversion before use.

3.5.9.12 Three-second fade logic. When the velocity of the target falls below 10 M/s, the radar shall maintain the track mode for 3.00 ~0.25 seconds after loss of doppler signal.

3.5.9.13 One-second fade logic. When the radar is tracking a target with a velocity between 10 M/s and 310 M/s and target return sigaal reduces to a value below the MDS of the radar, the radar shall maintain t-he track mode for 1.0 ~0.02 second after the signal level drops below the MDS of the radar with the range rate equal to the last value of range rate before loss of signal ~20 MIs2.

3.5.9.14 Range rate lag. The range rate lag, when tracking a target which is accelerating from 0 to 243 M/s, at an 81 M/s2 rate, shall not exceed 30 M/s.

3.5.9.15 Minimum range tracking. The radar shall track an inbound target to a minimum range of 250 meters before breaking track.

3.5.9.16 Receiver bandwidth. The 3 dB bandwidth of the radar receiver shall be 1.9 ~0.2 MHZ.

3.5.10 Antenna characteristics. The antenna shall meet all the requirements of MIL-A-50687.

3.6 First article. When s ecified in the contract or purchase order, a sample sha~e' sub ec @d to first article Inspection in accordance with the technics1 provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance a with the general requirements of MI&-sTD-2000 in addition to detail , , requirement (task)F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification-shall become a-part of the contractor's overall inspection system or quality program.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be Conducted at the following standard ambient conditions

a.	Temperature	+50 deg to +90 deg F.
b.	relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of H G

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three radar sets for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table S1 (Functional Test) and Table III (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimIen to comply with any of the stated requirements .

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term 'inspection lot" is defined as a homogeneous collection of units of product from which

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:31Z
Check the source to verify that this is the current version before use.

a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same **specifications and same specification revisions.** All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 **Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained i~ this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.**

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to government acceptance or rejection by item or lot.

4.4.2.3 **Functional tests. The tests in Table II shall be performed on a 100% basis.**

TABLE II. Functional tests
(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
101.	Electrical	3.4	4.'7.2
102.	Performance	3*5	4.7.3
Minor:	None	N/A	N/A

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:31Z
Check the source to verify that this is the current version before use.

4.5 Special sampling. One radar out of the first 50 produced and one out of each 100 produced thereafter shall be selected at r and om. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract the items shall be returned to the' lot upon successful completion of testing.

TABLE III. Special sampling

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3*3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105* - Altitude	3*3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the Last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment. Where 'the contract provides for Government-furnished test equipment. supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the 'Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product toler-

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:31Z
Check the source to verify that this is the current version before use.

ante for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used In the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IV. Test equipment list

ITEM NUMBER		DESCRIPTION
1.	Power Supply	0 to 36 V @ 30 A, regulation ± 0.005%, ripple less than 1 mV rms.
2.	Headset	H-140 A/u (cage: 80058)
3.	Oscilloscope	Dual trace, 150 MHz bandwidth, both channels are Identical. Rise time: 2.4 ns with or without probe. Vertical deflection: 5mv/div. dc to 60 MHz rise time: 5.9 ns; 10 mv div. dc to 100 MHz, 3.5 ns rise time. Calibrated deflection: from 5 mv/div. to 10 V/div. in 11 steps. Input RC: 1 megohm paralleled by 20 pF. Horizontal deflection: 0.05 us/div. to 5 s/div. in 25 steps. Triggering: dc to 150 MHz, x10 magnifier. Calibrated sweep delay: 1 us to 50 s. Sweep accuracy: ± 3%
4.	Waveguide rf load	Frequency range: 802 to 12.46 Hz. UG-39/U flange. VSWR : max. of 1.10 to 1. Maximum average power: 500 W, peak power not less than 3 KW.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE IV. Test equipment list (continued)

ITEM NUMBER	DESCRIPTION
5. Digital voltmeter	Dc voltage ranges 1, 10, 100, 1000 With 20% over-range on all ranges. Automatic polarity. Resolution of 0.01% of range (100 UV on 1 V range max.) Accuracy: ± 0.02% of input, +0.01% of range. Dc input resistance: constant 10 meg-ohms on all ranges. Overload protection: ± 1200 Vdc or ± 1700 V peak ac indefinitely on any range without damage.
6. Rf power meter	Automatic self balancing, 7 calibrated ranges from 10 UW to 10 MW and -20 to +10 dBm full scale in 5 dB steps, accuracy of ± 1% of full scale, external compensated thermistor mount required.
7. Thermistor mount	Frequency range: 10 MHz to 18 GHz. Maximum SWR: 1.35 to 1 from 1 to 12.4 GHz. Operating impedance: 200 ohms, 50-ohm match over frequency range. Temperature compensated 0.15 MW maximum input.
8. wave guide frequency meter	*Frequency range:* 8.2 to 12.4 GHZ. UG-39/U flanges. Overall accuracy: 0.08%. Dial increments: 5 MHz.
9* Waveguide coupler	Frequency range: 8.2 to 12.4 GHz. UG-39/U flanges. Coupling: 40 ± 0.3 dB. Coupling sensitivity: ± 0.5 dB max. Minimum directivity: 40 dB. Main line VSWR: 1.10 to 1 max. Secondary line VSWR: 1.20 to 1 max.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE IV. Test equipment list (continued)

ITEM NUMBER	DESCRIPTION
10. True rms voltmeter	Range: 1 mV to 300 V, 12 ranges. DB range: -72 to +52 dBm (600 ohms). Frequency range: 10 Hz to 10 MHz. Meter accuracy: ± 1% from 600 Hz to 10 KHz. Input impedance: 0.001 v to 0.3 v ranges, 10 megohms shunted by less than 50 pF, 1.0 V to 300 V ranges 10 megohms shunted by less than 20 pF. overload protection: 30 dB over full scale.
11. X-band argon tube noise source	Frequency range: 8.2 to 12.4 GHZ Excess noise *ratio:* 15.75 ± 0.5 dB. Anode current: 175 ma UG-39/A connector. Unfired SWR: 1.3 to 1, fired SWR: 1.2 to 1. Maximum unfired Insertion loss: 0.5 dB. Maximum fired insertion loss: 20 dB. Termination power rating: 4 W.
12. Noise source power supply	Adjustable output current: 0 to 250 mA. Separate controls for set and fire of noise source. Output current meter range: 0 to 250 mA
13. Waveguide variable attenuator	equal UG-39/U
14. Rf variable attenuator	
15. Dynamic range simulator	Target range: O - 999,999 ft with incremental resolution of 1 ft, accuracy of ± 1 ft. Range jitter: less than 2 *ns* +0.001% of range set. Target velocity: 0.0 - 99,999.9 ft/s with incremental resolution of 0.1 ft/s,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

	accuracy of ± 0.005% of instantaneous velocity. Target acceleration: 0.000 - 9,999*99 ft/s/s with Incremental resolution of O.O1 ft/s/s, accuracy of ± 0,005% of set acceleration. Reversals: Upon reaching zero range on any dynamic run, velocity direction automatically reverses; upon reaching zero velocity on any dynamic run, both velocity and acceleration automatically reverse. Trigger PRF: internal or external. Start and stop settings for range run: Director of range, change switch, reset switch, start-run switch, target fade switch . Target pulse output: 6 V min. into 50 k ohm equal to or less than 10 ns rise time, 400 ns width, positive polarity. External *trigger* input: 7 V peak max. Rep rate: 40 KHZ max, 0.33k-ohm Impedance, positive polarity.
16. Phase shifter	Center frequency: 60 MHz. Bandwidth: 10%. Phase shift range: O to 180 . Impededance: 5O ohms* Insertion loss. 1 db max. Manual control of phase shift.
17. Dual channel recorder	Variable" paper speeds: 1, 5, 20 and 100 mm/s. Frequency response: dc to 125 Hz. Range selection: V/cm or MV/mm of 0.5, 1, 2, 5, 10 or 20. Variable gain knob. Marker capability. NOTE : all specifications apply to each charnel.
18. X-band fixed attenuator	6 ± 0.3 dB over entire frequency range. Frequency range: dc to 12.4 GHZ. power capability: (max.) average 2 W, peak, 0.3 kW at duty cycle 0.001. Impedance: 50 ohms. VSWR : No worse than 1.3 to 1 across the band.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE Iv. Test equipment list (continued)

ITEM NUMBER	DESCRIPTION
19. Rf detector	Frequency range: 10 MHz to 12.4 GHz Frequency response: ±0.5 dB over full range. Maximum input power: 100 mW. Sensitivity with no load: input no greater than 0.4 mw to give 0.1 V rectified output. Square law response: within ± 0.5 dB from low level up to 1 mw input. VSWR: less than 1.4 to l to lO GHz; less than 1.5 to 1 from 1O to 12.4 GHz. Output capacitance: 30 UF. Video internal impedance: at O dbm input, 5 K ohms max; at -30 dBm input 20 K ohms max.
20. Precision dc power supply	Dc from O to 25 V at 500 mA, 4 decade stability and readout. Accuracy: ± 0.1% with 1 mV vernier. Stability: 100 UV for 8 hours. Resolution: 10 Uv Peak-to-peak ripple noise: 100 uv Source impedance: 0.20 milliohms Adjustable current limiting meter for V
21. Frequency counter	Dc coupled Q to 50 MHz. Gate time: 1 us to 10 s. Accuracy: ±1 count ± time base accuracy Readout: in KHz or MHz with positioned decimal point display, 8 digits in line. Adjustable trigger level. sensitivity: 100 mv rms. Sensitivity switch with ranges of 0.1, 1 and 10 V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE IV. Test equipment list (continued)

ITEM NUMBER	DESCRIPTION
22. Rf signal generator	Frequency range: 50 KHz to 65 MHZ. Accuracy: ± 1%, fine frequency vernier. Output level: continuously adjustable from 0.1 UV to 3 V into 50 ohms. Output attenuator: calibrated in 10 dB steps from 3 V full scale to 1.0 UV full scale, and also calibrated in dBm, with vernier control for continuous adjustment between 10 dB steps. Rf output meter: calibrated in dBm and volts . Auxiliary fixed level CW output jack minimum of 200 mV rms into 50 ohms from 19 to 65 MHZ.
23. Rf voltmeter	Frequency range: 20 KHz to 1.2 GHZ. Accuracy: ± 3%. Voltage sensitivity: 1 mV. DB range: 81 dB. Input impedance: greater than 100 K ohms. VSWR not more than 1.2 to 1; must Include 50-ohm termination matched to meter and/or probe.
24. Multimeter	Dc voltage ranges: 2.5, 10, 50 V. polarity reversal switch. Accuracy: ± 2% FS.
25. Rf test set	AN/VPM-2 Radar System Tester; 11748124, or equivalent.
26. System control unit	As specified in Figure 1. See Table V for fixture parts list.
27. Range information monitor	As specified in Figure 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE V. System control unit parts list

ITEM	DESCRIPTION
F1	Fuseholder and 1 A fuse
P1	Connector, MS31O6B16-105
P2	Connector, L22TF21S6N-A
K1	Relay, DPDT, 24 Vdc coil
Sl, S3	Switch, SPST, 2 Adc
S2	Switch, DPST, 5 Adc
DS1-DS3, DS5	LampS, 24 Vdc
DS4	Lamp, 115 V
Ml	Meter, running time: O to 1000 hours
P3	Connector, MS3126F-10-6P, or equivalent

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

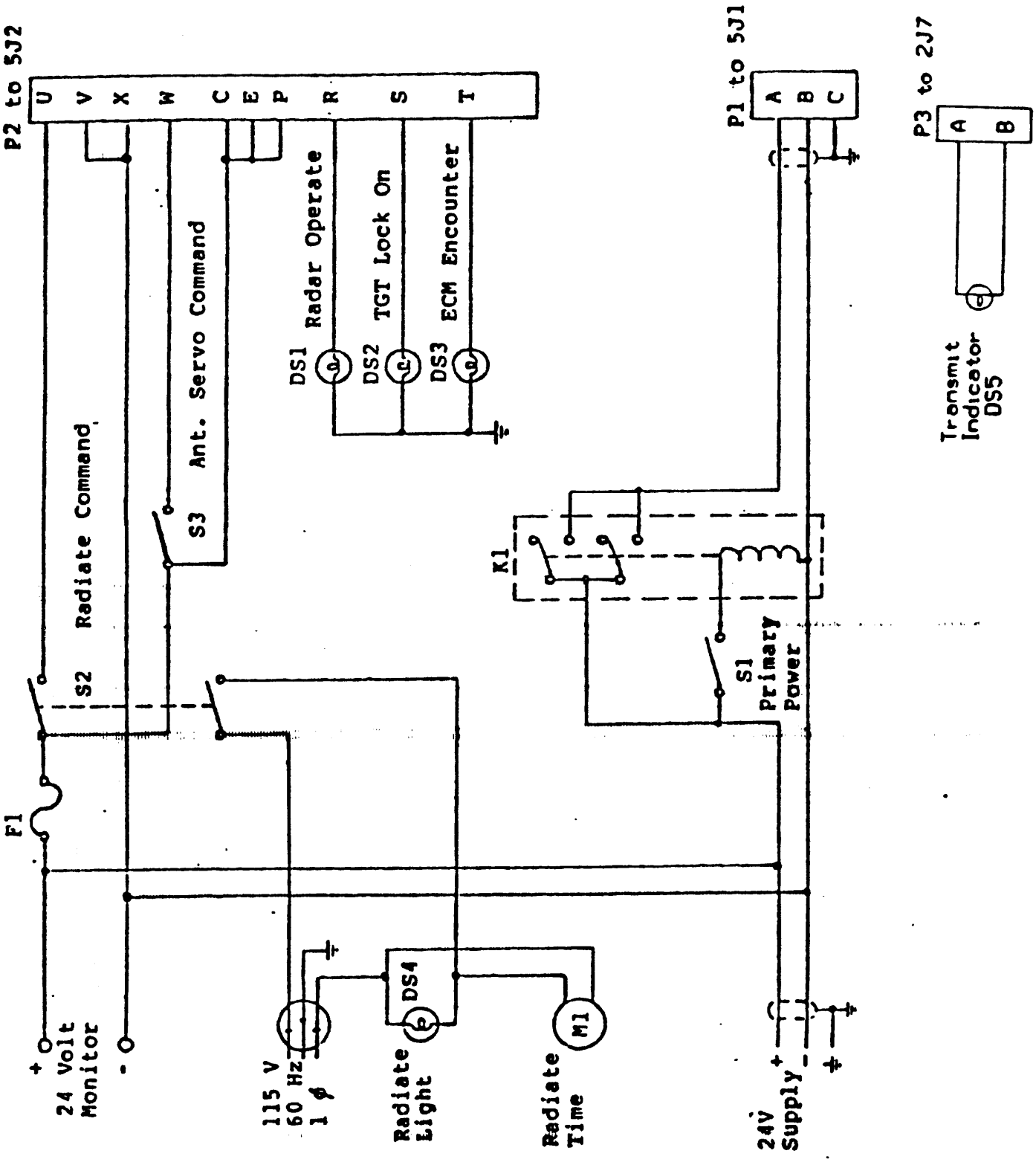
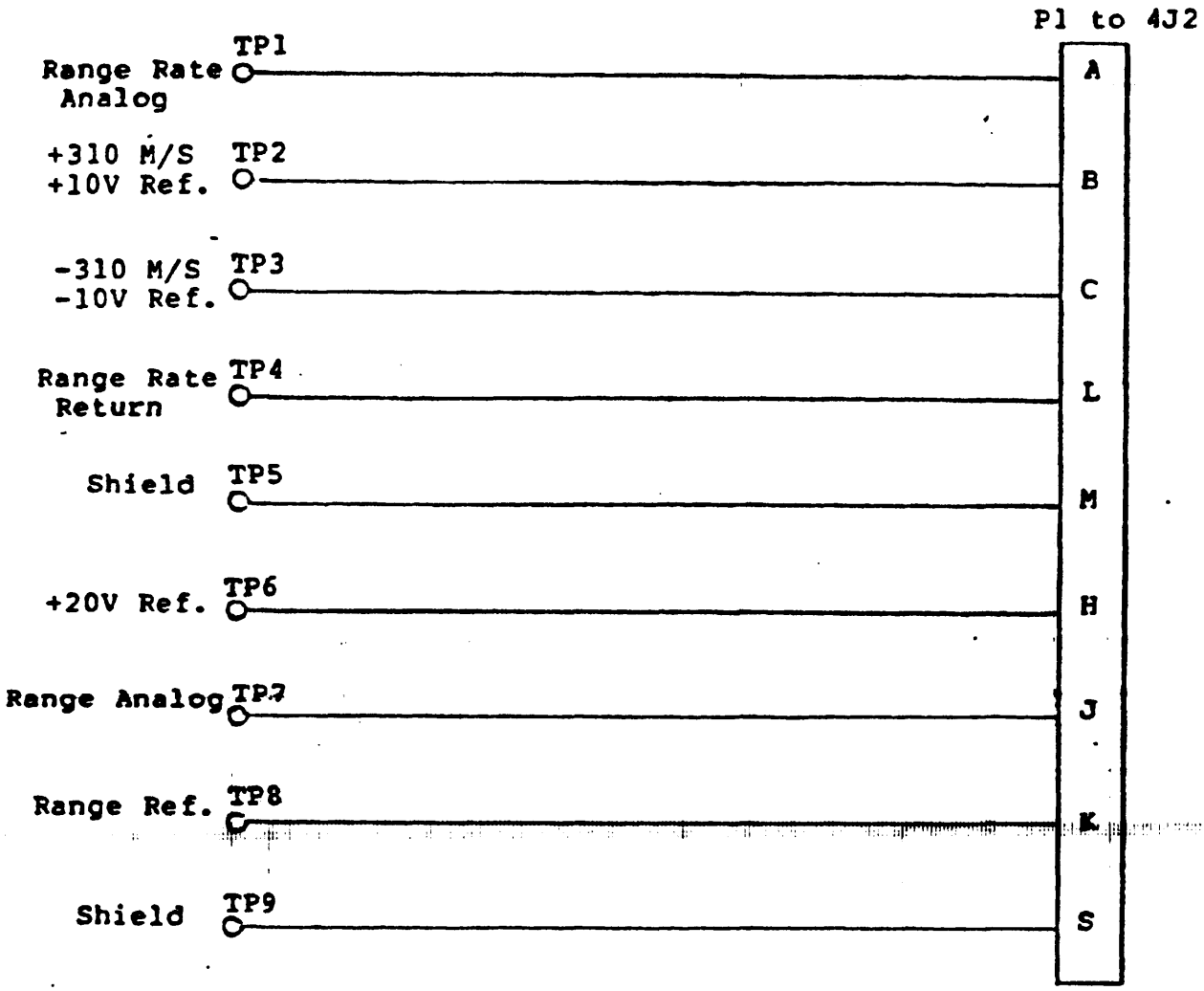


FIGURE 1. System control unit diagram.

Source: https://assist.dla.mil -- Downloaded: 20260-22T20:31Z
Check the source to verify that this is the current version before use.



NOTE: P1 - L20TF16P6N-A, Bendix.
TP1 through TP9, Meter Jack - H. H. Smith 1501 or equivalent.

FIGURE 2. Range information monitor.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:31Z
Check the source to verify that this is the current version before use.

4.7 **Methods of inspection.**

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 vibration. The test shall be performed as specified in Method 514, Category 8, Procedure 1, of MIL-STD-810 as modified by **3.3.1** herein. In Step 1 of procedure I only steps 4 through 8 of paragraph II-2.1 apply. The radar shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the radar shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4:7.1.2 Shock. This test shall be performed as specified in Procedure I, -on 11 of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. **The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.**

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz. After the application of all the shock impulses, the radar shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be Performed as specified in procedures II and III, Method **500** of **MIL-STD-810.** The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 **Hiqh (operating and non-operating) This test shall be performed as specified in procedures I and II, "Method 501 of MIL~STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.**

4.7.1.4.2 *LOW* **(operating and non-operating~ This test shall be performed as specified in Procedures I and II; Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only with the item at standard ambient conditions. The operational checkout shall be as specified In 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. The selected radars shall be tested as follows:

4.7.3.1 Initial test setup. Connect the radar and test equipment as shown in Figure 3 with the controls of units 2 and 4 of the units under test and the system control unit (Table IV, item 26) set as follows:

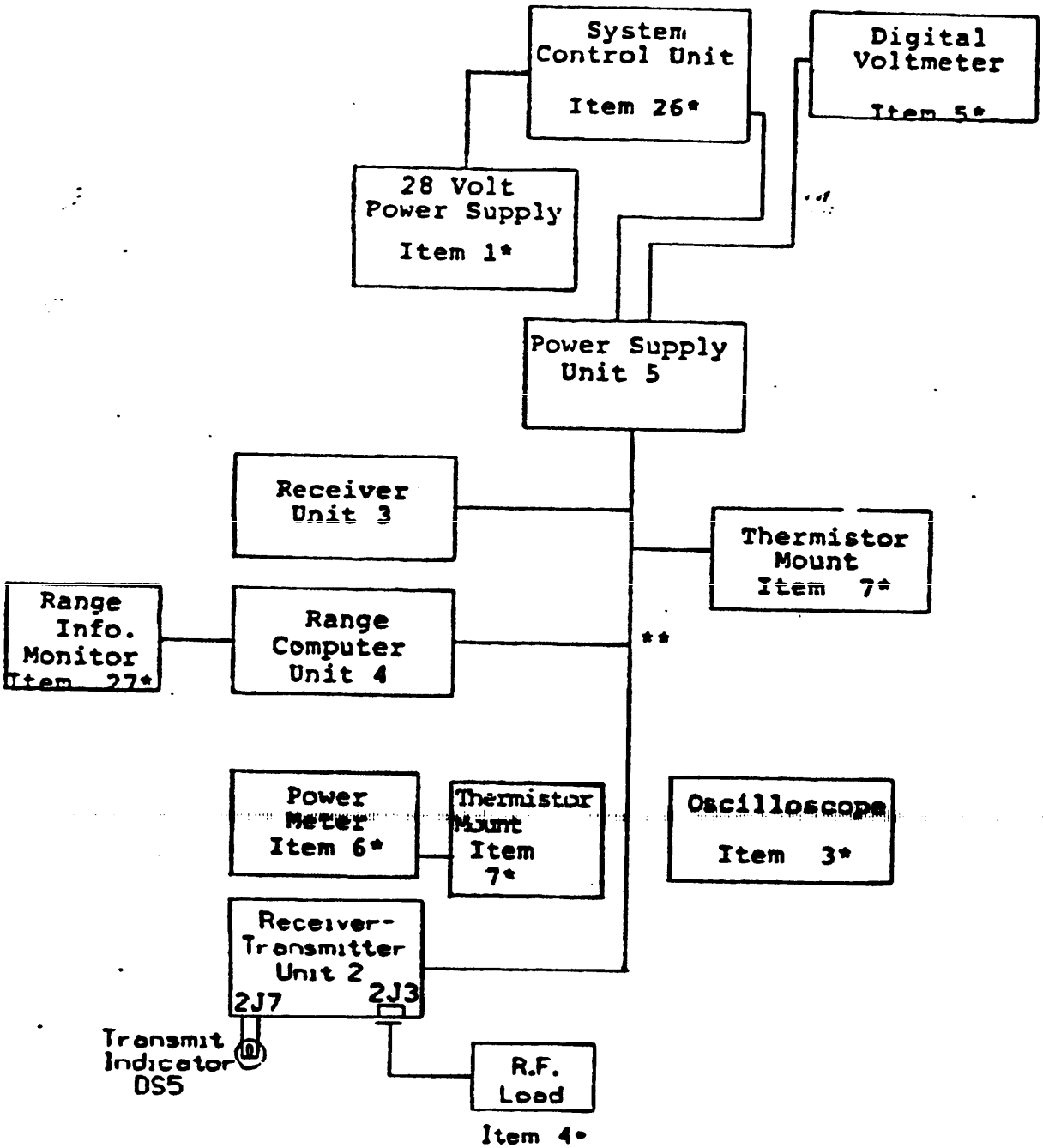
UNIT	UNITS UNDER TEST DESIGNATION	POSITION
2	S-2 CLUTTER LOCK ON SWITCH	NORMAL
2	S-3 DOOR INTERLOCK	PULL OUT
2	S-6 SER RADIATE CONTROL	OFF
4	S-2 MODE	NORMAL
4	S-3 ACCEL	POS.
4	S-4 RETURN	TGT
4	s-5 RANGE	RUN
4	S-6 RATE	RUN
5	CBl - CONVERTER CIRCUIT BREAKER	DEPRESSED
5	CB2 - TRAVERSE CIRCUIT BREAKER	OPEN
5	CB3 - ELEVATION CIRCUIT BREAKER	OPEN

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the currentversion before use.

UNIT	UNITS UNDER TEST DESIGNATION	POSITION
5	CB4 - CONTROL CIRCUIT BREAKER	DEPRESSED
5	CBS - BLOWERS	DEPRESSED
5	CB6 - 6.5 VDC	DEPRESSED
	SYSTEM CONTROL UNIT	
	DESIGNATION	
	S1 - PRIMARY POWER	OFF
	S2 - WiDIATE COMMAND	OFF
	S3 - ANT. SERVO COMMAND	OFF

With the digital voltmeter (VM) connected to +24 Vdc MON jacks on the system control unit (Table IV, item 28), adjust the power supply (Table IV, item 1) for +28.0 ± 0.1 Vdc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

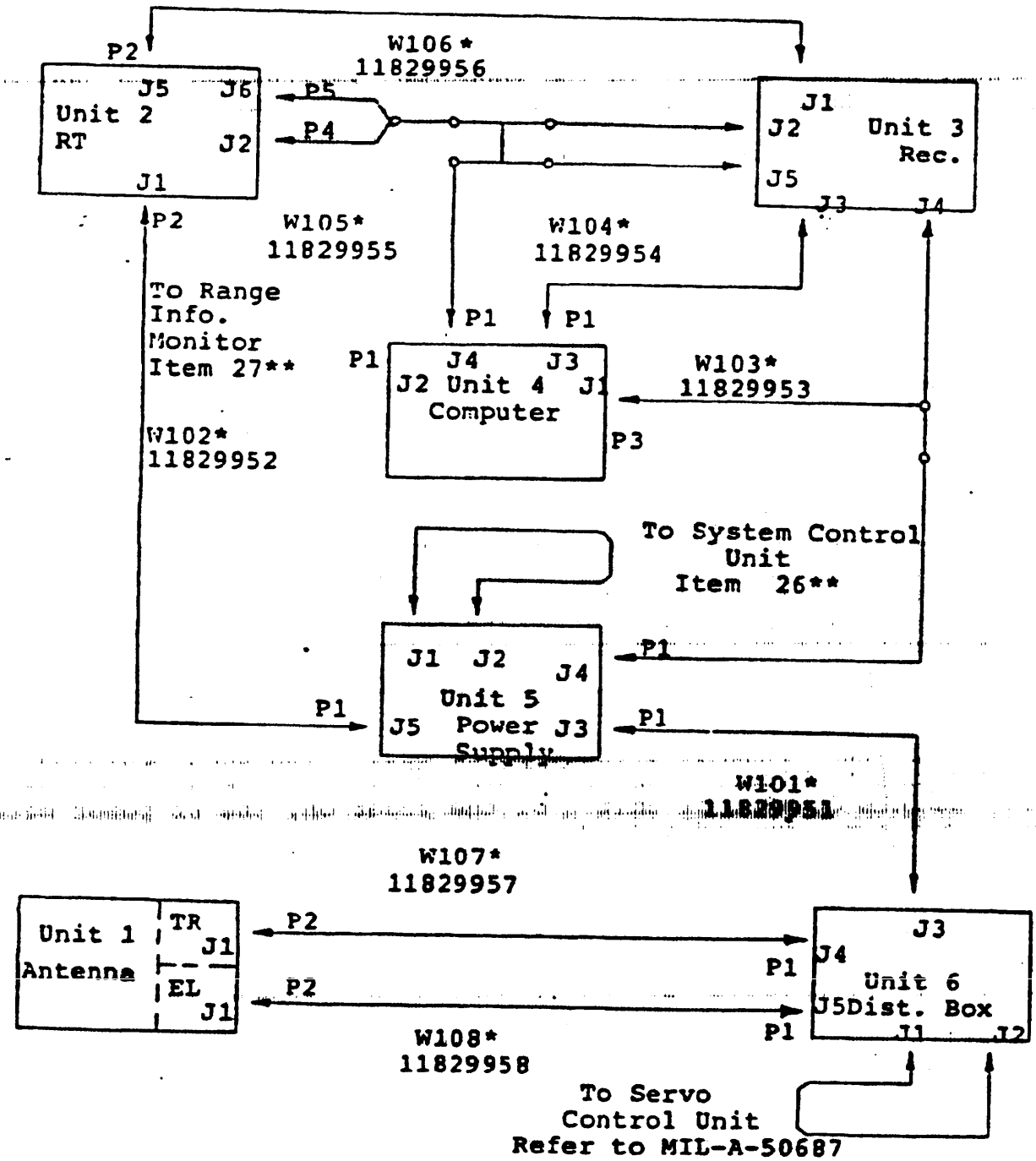


* Refer to 4.6.2.3

** System interconnecting cabling. See Figure 4.

FIGURE 3. Transmitting and receiving characteristics test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.



* Cable numbers listed to provide wiring information for system test cables.

** Refer to 4.6.2.3

FIGURE 4. System interconnecting cabling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

4.7.3.2 Standby power. With switch S1 PRIMARY POWER on the system control unit on, note time of the turn-on in order to monitor the time delay. In the meantime, verify that the fan in unit 2 and the fan in unit 5 are operating. Continue time monitor and note when the RADAR OPERATE light (+28 V between 5J2-R(+) and El(-) unit 5) on the system control unit lights. The total delay time must be 120 +30, -15 seconds.

4.7.3.3 **Transmit of indication and service radiate function. With the rf load (Table IV, item 4) connected to 2J3 of the radar, turn the SERV RADIATE CONTROL of unit 2 to the ON position. The TRANSMIT INDICATOR DS5 must light.**

4.7.3.4 Remote radiate function. With the RADIATE COMMAND switch of the system control unit on, the TRANSMIT INDICATOR DS5 must light.

4.7.3.5 **Unit 2 interlock check. With the lid on unit 2 box closed and screwed down, and the radiate command of the system unit on, the TRANSMIT INDICATOR DS5 must light. Turn radiate command switch OFF. With the lid on unit 2 box opened, and the radiate command of the system unit on , the TRANSMIT INDICATOR DS5 must be OFF . With the lid on Unit 2 box open, the interlock switch in manual overide position, and the radiate command of the system unit ON, the TRANSMIT INDICATOR Ds5 must light.**

4.7.3.6 Transmit characteristics.

4.7.3.6.1 Transmit average power out. With test equipment set up as shown in Figure 5 while retaining Figure 3 setup, and the SERV RADIATE CONTROL OF unit 2 of the radar on the RF power meter must read a minimum of 4.26 W.

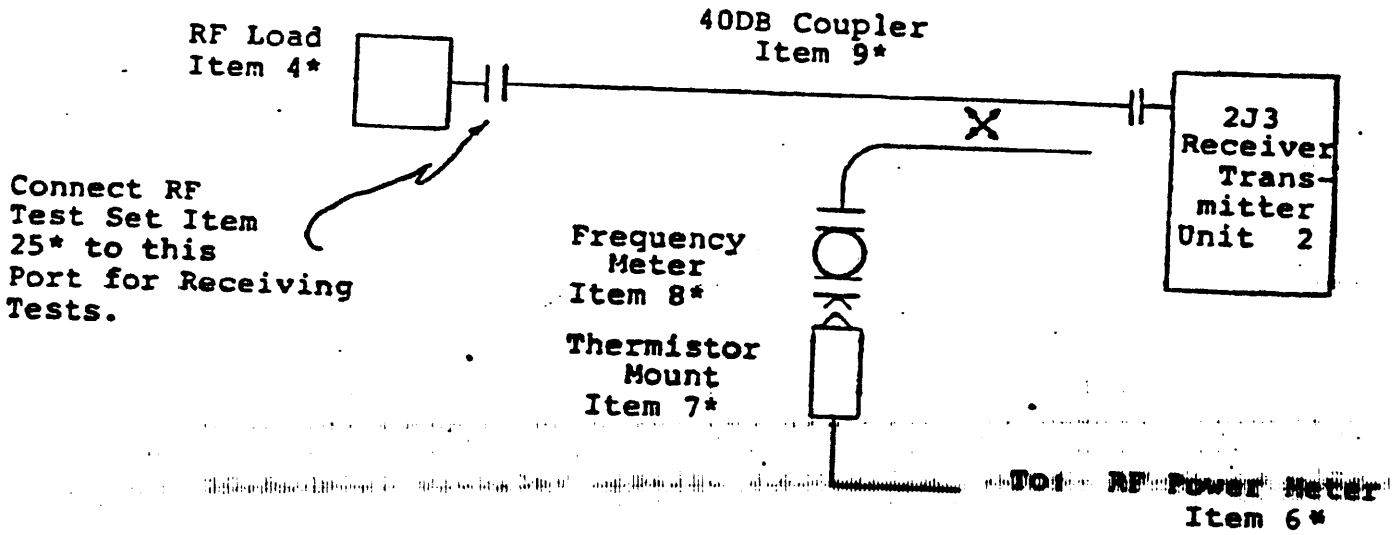
4.7.3.6.2 Transmit Pulse width. With the 6 dB attenuator (Table IV, item 18), and the X-band detector (Table IV. item 19) connected to the 40 dB coupler and a cable from the detector to the oscilloscope terminated in 50 ohms, the detected output pulse must be 280 ± 20 ns measured at the 50 percent amplitude points.

4.7.3.6.3 Transmit frequency. The frequency as read on the waveguide frequency meter (Table IV, item 8) must meet the requirements of 3.5.6.3.

4.7.3.6.4 Radar PRF. The PRF of the radar as read on the frequency counter (Table IV, item 21) must be 20.0.0.l Khz.

4.7.3.6.5 Remaining Channels. Repeat 4.7.3.6.1 through 4.7.3.6.4 for the five remaininq channels.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the current version before use.



*Refer to 4.6.2.3

FIGURE 5. Transmitting and receiving characteristics test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

4.7.3.7 Mid-range cal function.

4.7.3.7.1 Lock threshold setting.

a. Remove attenuator 3AT5 and replace with rf attenuator (Table IV, item 14).

b. Set the rf attenuator for 2 dB attenuation.

c. Adjust 4A15R22 ccw until the target lock on light on the system control panel comes on.

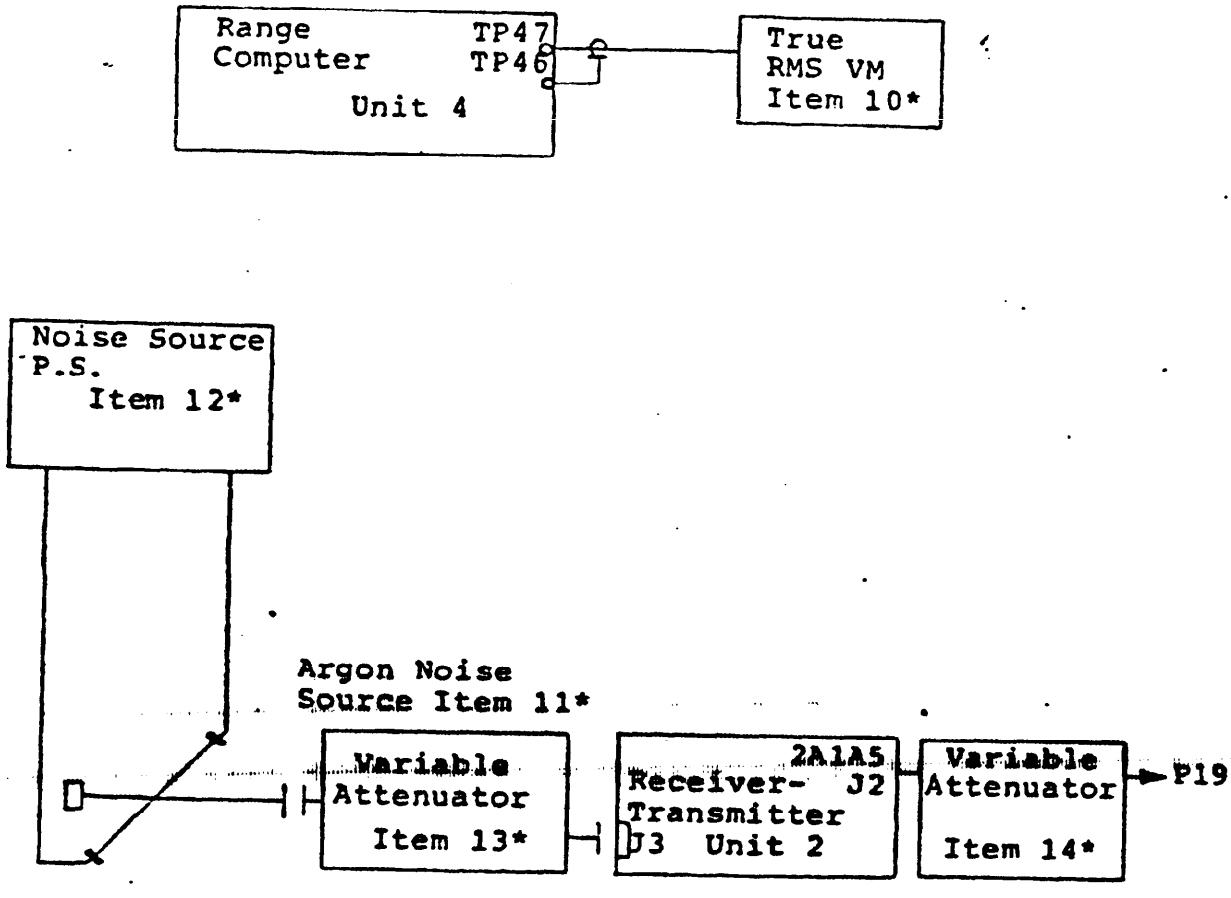
d. Adjust 4A15R22 cw until the target lock on light just goes out without flicker.

e. Remove the variable attenuator and re-install 3AT5.

4.7.3.7.2 Mid-range cal test. With the radar in the standby mode, -depress the mid-range calibrate button on unit 4 of the radar and observe that the mid-range cal light illuminates brightly. The range analog voltage at 4J2-J(+) and 4J2-K(-) should, concurrently, read 10.0 ± 0.2 Vdc.

4.7.3.8 System noise figure. With the test equipment connected as shown in Figure 6. the variable attenuator fTable IV, item 13) set for maximum attenuation, the variable attenuator (Table IV, item 14) set for O dB, the RANGE switch of unit 4 set to FREEZE, and the noise tube current set for 17s mA, note the noise voltage level on true rms voltmeter (Table XV, item 10). With variable attenuator (Table IV, item 14) set for 3 dB, reduce variable attenuator (Table IV, item 13) until noise voltage level is the same as nOted above. difference between variable attenuator reading and the excess noise ratio of argon tube noise source must be a noise figure of 13 dB maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.



*Refer to 4.6.2.3

FIGURE 6. Receiver noise test setup.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:31Z
Check the source to verify that this is the current version before use.

4.7.3.9 . The following self test procedure shall be acceptably performed on Rf test set (Table IV, item 25) before using it for receiver characteristics measurements.

4.7.3.9.1 AN/VPM-2 self test procedure.

a. voltage checks.

(1) Connect the equipment as shown in Figure 7. Do not connect flexible cable w5 at this time.

NOTE

Anytime the ANIVPM-2 system tester power is interrupted for more than 15 minutes voltage checks and self-checks should be repeated to ensure that the equipment is fully operational before continuing any tests.

(2) Set the AN/VPM-2 system tester controls as follows:

CONTROL	POSITION
POWER	OFF
FUNCTION SELECT	VOLTAGE
VOLTAGE SELECT	+24
RFLEVEL	20 dB
FREQ. SELECT	Radar transmit frequency (See Table VI.)
MODE SELECT	SELF TEST
PHASE CONTROL	MID RANGE
CLUTTER	OFF
CLUTTER LEVEL	20 dB
RANGE SELECT	R3
TARGET LEVEL	27 dB
TARGET FREEZE	OFF
TARGET	OFF

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

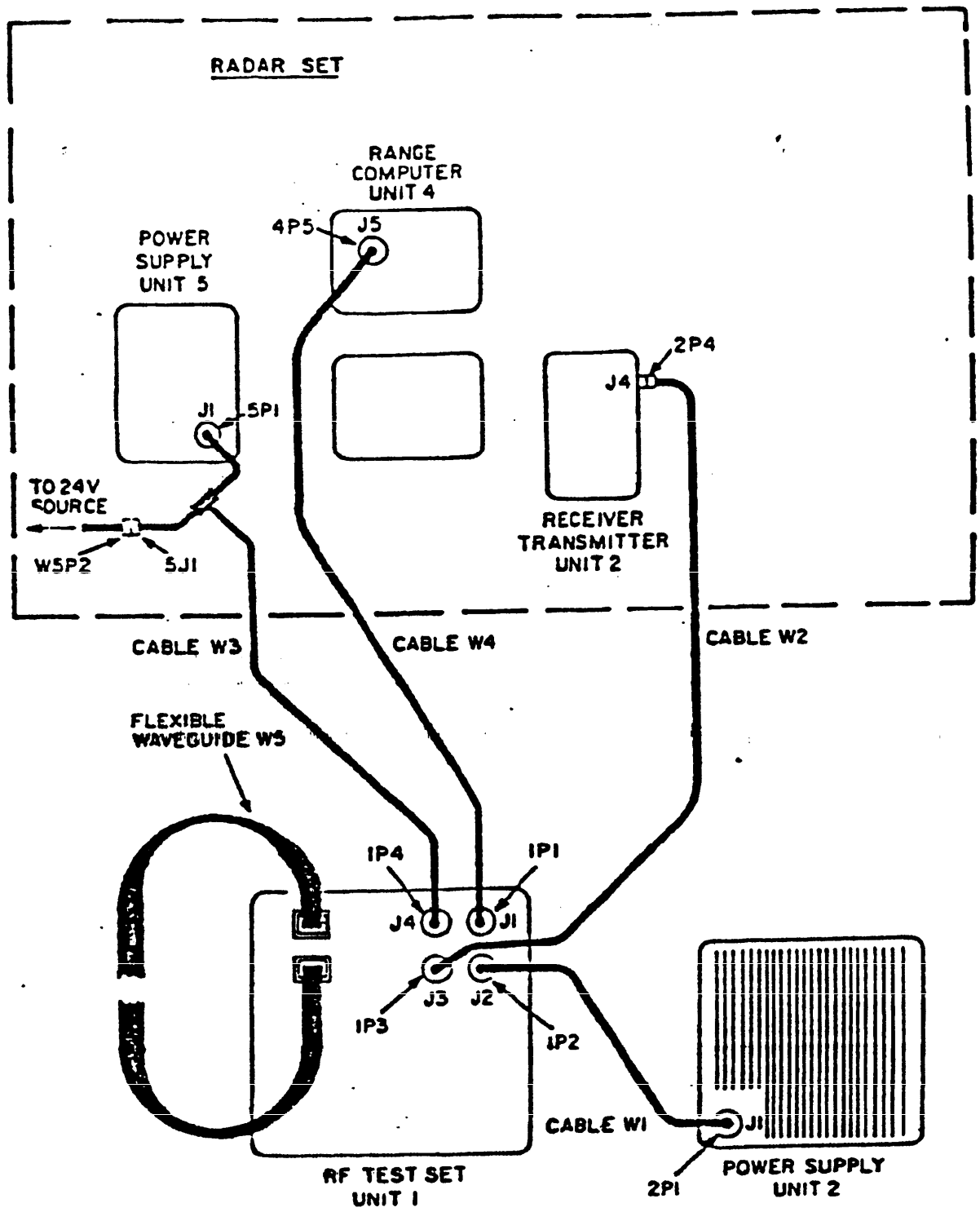


FIGURE 7. AN/VPM-2 system tester self test, equipment setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE VI. Radar Transmit Frequencies

TRANSMIT FREQUENCY (~Z~	STALO FREQUENCY GHz)	CHANNEL NO.
9.205	9.145	1
9.213	9.153	2
9.221	9.161	3
9.229	9.169	4
9.237	9.177	5
9.245	9.185	6

(3) On the AN/VPM-2 system tester, set POWER switch to ON. Check that the digital display is illuminated.

(4) observe that the digital display indicates +24.0 ± 2.o Vdc.

(5) Set the VOLTAGE SELECT switch to the following positions. Note the digital display at each setting.

VOLTAGE SELECT	DIGITAL DISPLAY (vdc)
VOLT TP	# * 0 ± o.o15
"+5	+ 5.0 ± 0.2
+15	+15.0 ± o.3
- 1 5	-1500 ± 0.3
+ 2 8	+28.0 ± 0.6

b. Self test.

(1) RF input.

NOTE

Be sure that the pin on the flexible waveguide coupling flange is properly keyed to the RADAR port. Be sure that both waveguide sections are properly aligned.

(a) Connect flexible cable W5 between the RADAR port and the SELF TEST port of the AWVPM-2 system tester.

(b) Set the FUNCTION SELECT switch to R. F. INPUT.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the currentversion before use.

(c) Adjust the RF LEVEL control to obtain a digital display reading of 0.500 ± 0.075 Vdc. The RF LEVEL control setting should be greater than 3.o *(db).*

(d) Set the FUNCTION SELECT switch to R.F. MONITOR.

(e) Adjust the RF LEVEL control to obtain a digital display reading of +2.50 ±.075 Vdc.

(f) Vary the PHASE CONTROL over its entire range. The digital display should remain between +2.o and 3.o Vdc. Reset the PHASE CONTROL to mid-range.

(g) Adjust the RF LEVEL control to obtain a digital display reading of +2.50 ± 0.075 Vdc.

(h) Set the RANGE SELECT switch to Rl, R2, and R4 . The digital display should remain between +1.30 and +3.7o vdc.

(i) *Set* the RANGE SELECT switch to R3. Set the CLUTTER switch to ON. Set the CLUTTER LEVEL control to 20 dB.

(j) Adjust the RF LEVEL control for a digital display reading of 2.50 ± 0.075 Vdc.

NOTE

the 2.50 +0.075 vdc reading is a reference level injected into the target and clutter channels. Monitor and maintain this voltage throughout the procedure unless instructed otherwise.

(k) Set the FUNCTION SELECT switch to R.F. SELF TEST .

(1) Adjust the CLUTTER BALANCE for a digital display within ± 0.1 Vdc of the value indicated on the AWVPM-2 system tester calibration chart for *RF* self test.

(m) Set the FUNCTION SELECT switch to RF MONITOR. Repeat step (j) above.

(n) Set the FUNCTION SELECT switch to RF SELF TEST.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the current version before use.

(o) Set the CLUTTER LEVEL control to 25 dB.

(P) Adjust the RF LEVEL control to obtain a digital display reading of +0.750 ± 0.250 Vdc.

[q) Set CLUTTER switch to OFF. DVM should ind-. cate less than +0.075 Vdc.

(r) Set the TARGET switch to ON and the TARGET LEVEL control to 27 dB.

(s) Set the TARGET switch to OFF. The digital display should indicate less than +0.075 Vdc.

(t) Set the CLUTTER switch to ON and note the reading on the digital display.

(u) Set the CLUTTER switch to OFF and the TARGET switch to ON.

(v) Adjust the TARGET BALANCE control for the same digital display reading obtained in step (t) above within &O.025 Vdc. Set TARGET switch to OFF.

(w) Repeat steps (t) through (v) above until the digital display indications in steps (t) and (v) are within +0.025 Vdc. Do not readjust the CLUTTER BALANCE control.

(x) Set the CLUTTER and TARGET switches to OFF.

(2) RF Level attenuator.

(a) Set the FUNCTION SELECT to RF MONITOR.

(b) Adjust the RF LEVEL control for a reading of +3.16 ± 0.025 Vdc on the digital display. Record the RF LEVEL control setting.

(c) Increase (higher number on dial) the RF LEVEL setting by 5.0 dB. The digital display should indicate 1.00 ± 0.25 Vdc.

(3) CLUTTER LEVEL attenuator.

(a) Set the FUNCTION SELECT to RF MONITOR.

(b) Set the CLUTTER switch to ON.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

(c) Adjust the RF LEVEL control for a reading of 3.0 ± 0.07s Vdc on the digital display. Record the RF LEVEL control setting.

(d) Set the FUNCTION SELECT to RF SELF TEST.

(e) Adjust the CLUTTER LEVEL control for a reading of +3.o ± 0.075 Vdc on the digital display. Record the CLUTTER LEVEL control setting.

(f) Increase (higher number of dial) the RF LEVEL setting by 5.0 dB.

(9) Decrease (lower number on dial) the CLUTTER LEVEL control to indicate 3.0 ± 0.075 Vdc on the digital display. Record the CLUTTER LEVEL control setting.

(h) Subtract the CLUTTER LEVEL control setting in step (g) above from the CLUTTER LEVEL control setting in step (e) above. The reading should be 5.0 ± 0.6.

(1) Repeat steps (f) through (h) above in 5.0 dB increments. The last increment may be less than 5.0 dB.

(j) Set the CLUTTER switch to OFF.

(4) TARGET LEVEL attenuator.

(a) Set the FUNCTION SELECT switch to RF MONITOR. Set the CLUTTER switch to ON.

(b) Adjust the RF LEVEL control for a reading of +3.0 ± 0.075 Vdc on the digital display. Record the RF LEVEL control setting.

(c) Set the FUNCTION SELECT to RF SELF TEST, the CLUTTER LEVEL control to 25 dB, and the TARGET LEVEL control to 27 dB (fully cw).

(d) Increase (higher number on dial) the CLUTTER LEVEL setting by 5 dB. Record the digital display indication.

(e) Set the CLUTTER switch to OFF and the TARGET switch to ON.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

(f) Increase (higher number on dial) the TARGET LEVEL control for the same DVM indication (within +0.075 Vdc) obtained in step (d) above. Subtract 27 (original setting of step [c] above) from this setting. The result should be 5.0 ± 0.6.

NOTE

Self test of the AN/VPM-2 system tester has now been completed for the radar transmitter frequency selected. If another transmitter frequency is selected, a complete self test of the system tester must be conducted for that frequency.

(g) Discontinue the flexible waveguide from SELF TEST port, and replace SELF TEST cover.

4.7.3.9.2 Receiver test.

a. Control settings.

(1) Set the AN/VPM-2 radar system tester controls as follows:

CONTROL	POSITION
FUNCTION SELECT	VOLTAGE
VOLTAGE SELECT	+24
RF LEVEL	20 dB
FREQ SELECT	Radar transmit frequency (refer to Table VI).
MODE SELECT	RADAR
PHASE-CONTROL	Mid range
CLUTTER	OFF
CLUTTER LEVEL	80 dB
RANGE SELECT	R3
TARGET LEVEL	100 dB
TARGET FREEZE	OFF
TARGET	OFF
POWER	ON

b. Waveguide W5 connection.

(1) DISCONNECT the rotary jOint of the antenna, unit 1, by removing 4 screws, 4 flat washers, and 4 lock washers from the waveguide section as shown in Figure 8.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:31Z
Check the source to verify that this is the current version before use.

(2) Push the waveguide section away from the rotary joint being careful not to destroy or loosen the rf seal located between the rotary joint and waveguide section. Connect the 90 degree waveguide from the accessory kit to W5 and connect W5 to the rotary joint as shown in Figures 8 and 9.

CAUTION

Ensure that waveguide W5 is connected between the radar and AN/VPM-2 system tester for proper termination to prevent damage to equipment.

c. **Transmit frequency determination.** Open cover of radar transmitter unit 2 and determine radar transmit frequency by adding 60 MHz to the frequency marked on solid state crystal of X-Band source (refer to Table VI).

d. **Radar turn-on preliminary.**

(1) **Turn ON the System POWER switch.**

(2) **Pull the service interlock switch S3 to the Maintenance Position.**

(3) **Wait a minimum of 120 +30, -15 seconds and then set the SERV RAD CONT switch located on the inside of the top cover of R/T unit 2 to ON.**

(4) **The digital display should indicate +24.0 ± 2.0 VDC**

(5) Set **the VOLTAGE SELECT to the following positions and observe the digital display.**

VOLTAGE SELECT	DIGITAL DISPLAY (Vdc)
+5	+5.0 ± o.2
+15	+15.0 ± o.3
-15	-15.0 ± o.3
+28	+28.0 ± 0.6

(6) **Ensure that the LOCK ON indicator is not illuminated and the SIGNAL CHANNEL meter indicates less than 1.0 Vdc.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

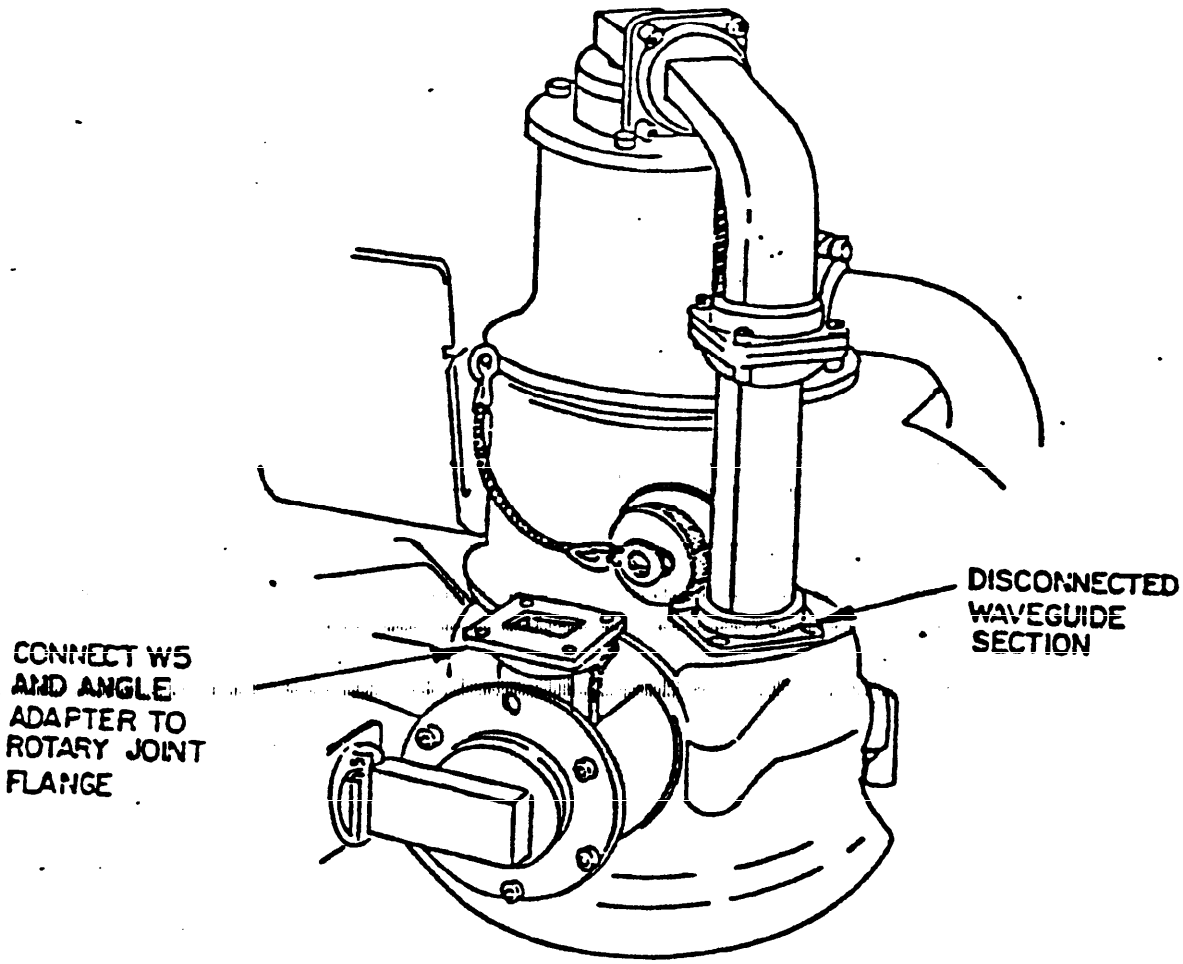


FIGURE 8. Antenna unit 1, rotary joint disconnection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

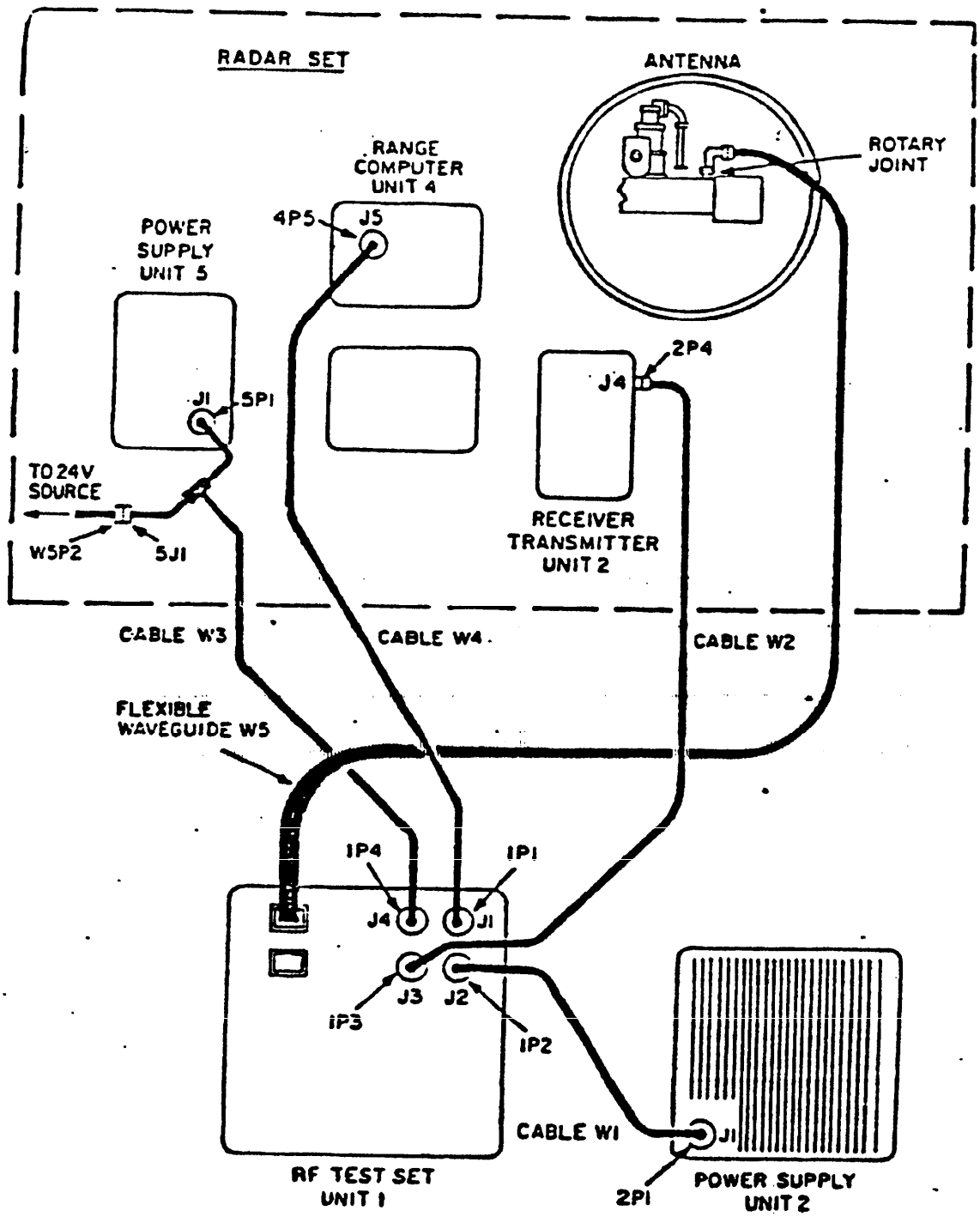


FIGURE 9. AN/VPM-2 System tester RADAR TEST equipment setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

4.7 .3.9.3 Target lock-on indicator and clutter lock-on. With the radar in the radiate mode and TARGET switch on, the TARGET LEVEL attenuator should be adjusted until the TGT LOCK ON light on system control unit (Table IV, item 26) lights without flicker (+28 V between 5J2-5(+) and el(-) on unit 5). Then the S5 RANGE switch of unit 4 of the radar is put in the FREEZE position, the AN/VPM-2 TARGET switch is turned OFF, and the CLUTTER switch ON. set CLUTTER LEVEL attenuator to 80 dB and clutter lock-on switch on unit 2 to test position. With headset (Table IV, item 28) connected to the DOPPLER AUDIO test jack on unit 2 of the radar, there must be a doppler audio signal clearly audible in the headset.

NOTE

For the following voltage checks, ensure that the TARGET and CLUTTER switches are set to OFF.

4.7.3.9.4 Range reference voltage. Set the FUNCTION SELECT switch to RANGE REF. The digital VM reading must be +20.0 ± 0.2 Vdc.

4.7.3.9.5 Range standby voltage. Set the FUNCTION SELECT switch to RANGE. The digital VM reading must be +6.4 ± 0.1 Vdc.

[b] 4.7.3.9.6. Range rate positive reference voltage. Set the FUNCTION SELECT switch to RANGE RATE REF (+). The digital VM reading must be +10.0 ± 0.2 Vdc.

4.7.3.9.7 Range rate negative reference voltage. Set the FUNCTION SELECT switch to RANGE RATE REF (-). The digital VM reading must be -10.0 +0.2 Vdc.

4.7.3.9.8 Range rate standby voltage. Set the FUNCTION SELECT switch to RANGE RATE. The digital VM reading must be -6.4 ± 0.1 vdc.

4.7.3.9.9 Range accuracy measurement Procedure.

a. Set the FUNCTION SELECT switch to RF MONITOR and the RANGE SELECT switch to R1.

b. Adjust the RF LEVEL control to obtain a reading of +2.500 ± 0.075 Vdc on the digital display.

c* Set the TARGET switch to ON.

d. Adjust the TARGET LEVEL control until the LOCK ON indicator illuminates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

e. Decrease (lower number on dial) the TARGET LEVEL control setting by 10 dB.

f. Set the FUNCTION SELECT switch to RANGE.

g* Adjust the PHASE CONTROL for a maximum stable indication on the digital display.

h. Read and record the range voltage indicated on the digital display.

NOTE

Allow 3 seconds for the radar to re-acquire the target signal each time the RANGE SELECT switch is changed.

i. Set the RANGE SELECT switch to R2, R3 and R4 and repeat steps g. and h. above.

j. Determine the range of the radar set using the following procedure:

(1) Using Figure 10, locate the voltage (step h. above) for R1 and record the corresponding range in meters.

(2) USing Figure 11 locate the range reference voltage [4.7.3.9.4) and record the correction for R1.

(3) Add the results of the two preceding steps. This is the measured range of the radar in R1 position. Record this number.

(4) Record the front panel AN/VPM-2 calibration number for R1.

(5) Determine the radar range error by finding the difference between the AN/VPM-2 calibration card data and the corrected range.

(6) Repeat steps (1) through (5) above for remaining range positions R2, R3, and R4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

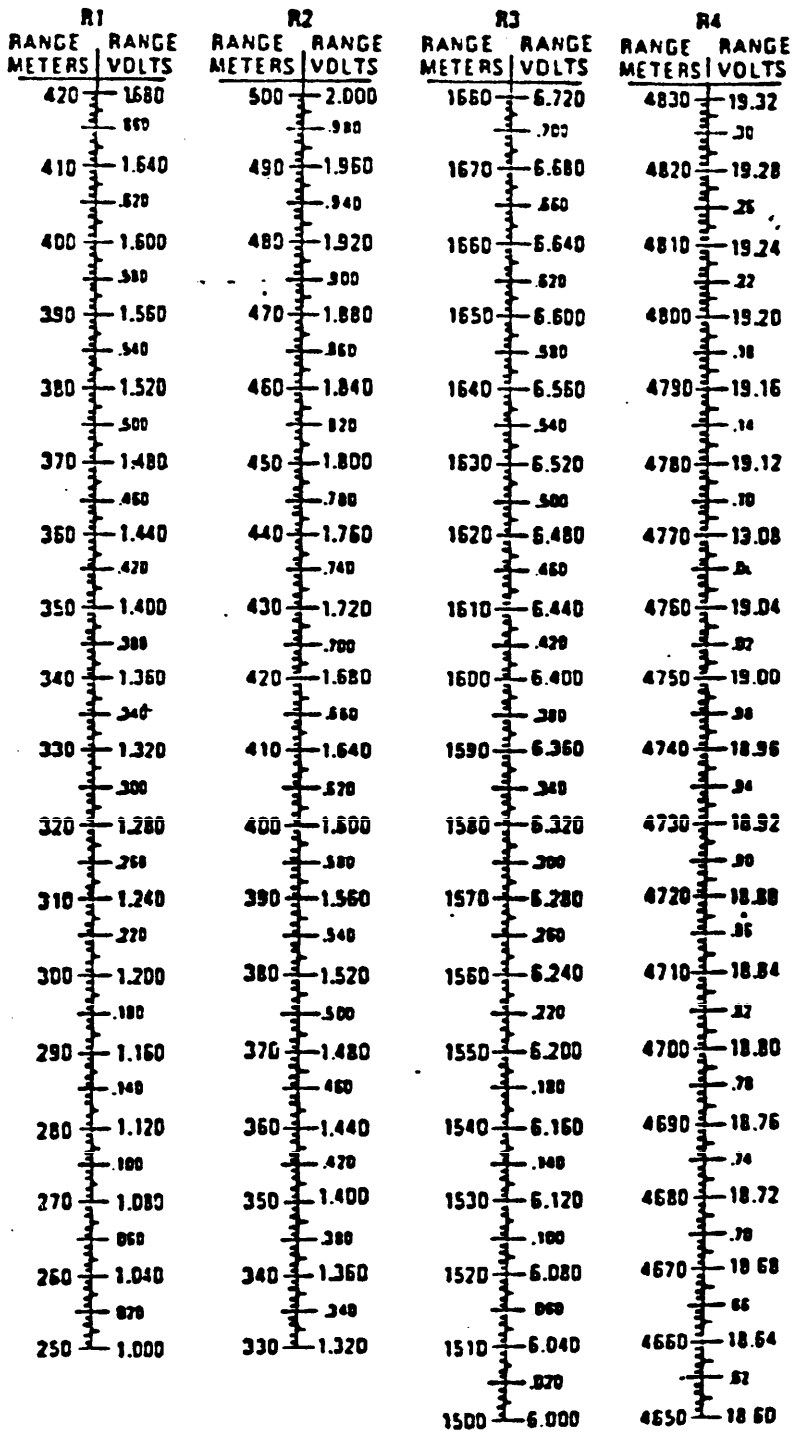


FIGURE 10. Meters to volts conversion chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

Range Reference Voltage	R1	R2	R3	R4
20.20	-3	-4	-16	-47
20.19	-3	-4	-15	-45
20.18	-3	-4	-14	-43
20.17	-3	-3	-14	-40
20.16	-2	-3	-13	-38
20.15	-2	-3	-12	-35
20.14	-2	-3	-11	-33
20.13	-2	-3	-10	-31
20.12	-2	-2	-9	-28
20.11	-2	-2	-9	-26
20.10	-2	-2	-8	-24
20.09	-1	-2	-7	-21
20.08	-1	-2	-6	-18
20.07	-1	-1	-6	-17
20.06	-1	-1	-5	-14
20.05	-1	-1	-4	-12
20.04	-1	-1	-3	-9
20.03	0	-1	-2	-7
20.02	0	0	-2	-5
20.01	0	0	-1	-2
20.00	0	0	0	0
19.99	0	0	+1	+2
19.98	0	0	+2	+5
19.97	0	+1	+2	+7
19.96	+1	+1	+3	+9
19.95	+1	+1	+4	+12
19.94	+1	+1	+5	+14
19.93	+1	+1	+6	+17
19.92	+1	+2	+6	+19
19.91	+1	+2	+7	+21
19.90	+2	+2	+8	+24
19.89	+2	+2	+9	+26
19.88	+2	+2	+9	+28
19.87	+2	+3	+10	+31
19.86	+2	+3	+11	+33
19.85	+2	+3	+12	+35
19.84	+2	+3	+13	+38
19.83	+3	+3	+13	+40
19.82	+3	+4	+14	+43
19.81	+3	+4	+15	+45
19.80	+3	+4	+16	+47

FIGURE 11. Range reference correction factors (in meters).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

(7) The difference between the calculated range and the value indicated on the AN/VPM-2 calibration card must fall within the following tolerances:

RANGE	TOLERANCE
R1	± 15 meters
R2	± 15 meters
R3	± 16 meters
R4	±47 meters

4.7.3.9.10 SCV without AGC. First perform the Minimum Discernible Signal (MDS) test (a) and then the Sub-clutter Visibility (SCV) measurement procedure (b). In both cases, operate the radiate switch to radiate. Preset the CLUTTER LEVEL dial to 80 dB and the CLUTTER switch to OFF.

a. Minimum Discernible Signal (MDS).

(1) Set the TARGET switch to ON and FUNCTION SELECT to RF MONITOR.

(2) Adjust the RF LEVEL control and maintain a digital display indication of +2.5 ± 0.075 Vdc for the remainder of this procedure, step a.

(3) Decrease (lower number on dial) the TARGET LEVEL control until the LOCK ON Indicator illuminates. Note the TARGET LEVEL dial indication.

(4) Increase (higher number on dial) the TARGET LEVEL b 10.0 dB. The LOCK ON indicator should not illuminate after an approximate 3 Second delay.

(5) Repeat steps (3) and (4) above until the last reading obtained in step (3) above is within 1.0 dB of the previous reading. Record this last reading. This is the radar system MDS.

NOTE

The calculated MDS acceptability criteria is 80-88 dBm.

(6) Set the TARGET switch to OFF and the TARGET LEVEL control to 100 dB.

b. Sub-clutter visibility (SCV) Measurement Procedure.

(1) Set the RANGE SELECT switch to R4.

(2) Connect the TPS shorting plug (P/O accessory kit) to J6 of receiver, unit 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

(3) **Set** the **FUNCTION SELECT switch to RF MONITOR.**

(4) Set the CLUTTER switch to ON.

(5) Check the RF LEVEL control adjustment and maintain a reading of 2.500 ± 0.075 Vdc on the digital display for the remainder of this procedure step b.

(6) Adjust the CLUTTER LEVEL control until the LOCK ON indicator is illuminated. Note the CLUTTER LEVEL dial reading. (Light does not have to stay locked on.)

(7) Increase (higher number on dial) the CLUTTER LEVEL control setting by 10.0 dB. The LOCK ON indicator should not be illuminated after an approximate 3-second delay.

(8) Carefully repeat steps (6) and (7) above until a reading obtained in step (6) above is within 1.0 dB of the previous reading.

(9) Record the CLUTTER LEVEL dial reading obtained in step (8) above. This is the lock on clutter power (no AGC).

(10) Determine the SCV by subtracting the lock on clutter power obtained in step (9) (no AGC) from the minimum discernible signal (MDS) recorded in step a. (5) above. The SCV acceptability criteria is 40 dB minimum.

(11) Repeat steps b.(l) through b. (10) *for* **the remaining three target ranges.**

(12) Repeat steps b. (l) through b.(n) for the remaining five channels of the radar.

(13) Set the **CLUTTER** switch to **OFF**, the **CLUTTER LEVEL control to 80 dB, and the RANGE SELECT switch to** R1 . **Remove the TPS shorting plug from J6 of receiver, unit 3, and return it to the accessory kit.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

4.703.9.11 .

a. Linearity.

(1) Set the CLUTTER LEVEL control to 80 dB and the TARGET LEVEL control to 100 dB.

(2) Set the CLUTTER switch ON.

(3) Set the FUNCTION SELECT switch to R.F. MONITOR. Adjust the RF LEVEL control and maintain a digital display indication of 2.500 ± 0.075 Vdc for the remainder of this linearity check.

(4) Decrease the CLUTTER LEVEL control setting slowly to 5 dB while observing that the LOCK ON indicator does not illuminate.

(5) Set the CLUTTER switch to OFF and the TARGET switch to ON.

(6) Decrease (lower number on dial) the TARGET LEVEL control until the LOCK ON indicator illuminates. (Note: This is the MDS level).

(7) Set the CLUTTER switch to ON. The LOCK ON indicator should not be illuminated after approximately a 3 second delay.

(8) Increase (higher number on dial) the CLUTTER LEVEL control setting until the LOCK ON indicator is illuminated.

(9) Decrease (lower number on dial) the CLUTTER LEVEL setting by 10.0 dB. The LOCK ON indicator should not be illuminated after approximately a 3 second delay.

(lo) Increase (higher number on dial) the CLUTTER LEVEL control until the LOCK ON indicator is illuminated. Note the CLUTTER LEVEL dial setting at the lock-on point.

(11) Carefully repeat steps (9) and (10) above until the readings obtained in step (10) are within 1.0 dB .

(12) Record the CLUTTER LEVEL dial setting obtained in step (10) above. This is the LOCK ON CLUTTER POWER (with AGC).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

(13) Determine the AGC cut-in points by subtracting the LOCK ON CLUTTER POWER recorded in step (12) from the Minimum Discernible Signal (MDS) recorded in step 4.6.3.9.10a.(5). The AGC cut-in acceptability criteria is 38 to 40 dB.

(14) **Decrease (lower number on dial) the CLUTTER LEVEL control setting for three 10 dB decrements noting the three correspondingly decreased TARGET LEVEL settings required to re-establish LOCK ON each time. The allowable TARGET LEVEL tolerances are as follows:**

	CLUTTER LEVEL Dial		TARGET LEVEL Dial
1.	*L.O.C.P. -10 dB	results in	MDS - (9 to 14 dBm)
2.	present clutter setting-lo dB	results in	present target setting- (9 to 16 dBm)
3*	present clutter setting-10 dB	results in	present target setting- (9 to 18 dBm)

*L.O.C.P means lock on clutteRPOWER (with AGC)

(15) "Set the TARGET and CLUTTER channel switches to OFF.

4.7.3.9.12 SCV with AGC.

a. Set the TARGET LEVEL control to 88 DB and set the CLUTTER LEVEL control to 40 dB. Place the radiate switch in the radiate position. If the radar does not look onto the *target,* decrease (lower number on dial) the TARGET LEVEL control in 1 dB steps until three successive solid locks can be achieved. The TARGET LEVEL attenuation shall exceed the CLUTTER LEVEL attenuation by not less than 40 dB.

b. Repeat step a. with initial setting of TARGET LEVEL at 57 dB and CLUTTER LEVEL at 15 dB. The TARGET LEVEL attenuation shall exceed the CLUTTER LEVEL attenuation by not less than 40 dB.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the currentversion before use.

4.7.3.9.13 Active ECM indication.

a. Turn OFF the CLUTTER switch and adjust the TARGET LEVEL control until the radar ***just*** **acquires** the target. **Repeat for three successive locks and note the TARGET LEVEL control setting.**

b. With the radar locked onto the target, decrease (lower number on dial) the TARGET LEVEL control until the active ECM lamp on the system control panel illuminates. The TARGET LEVEL control setting shall be not less than 15 dB and not **more than 25 dB lower than the TARGET LEVEL control setting noted in step a.**

c. **Set the TARGET switch to OFF.**

d. **Take the radar out of the radiate condition.**

4.7.3.9.14 Range rate accuracy. **With the test equipment set up as shown in Figures 3, 5, 9, and 12, the radar in standby mode and the Function Select switch on the AN/VPM-2 set to the RANGE RATE REF (+) position, refer to Table VII. Program the simulator in accordance with RUN 1. Press the RESET button on the simulator. Adjust the variable attenuator (Table IV, item 14) for a 60() mV peak-to-peak reading on the scope. Adjust the phase shifter (Table IV, item 16) to peak this signal, then readjust the attenuator for 600 mV peak-to-peak. Press the START button of the simulator. The RANGE RATE SIGNAL analog voltage ratio to the proper range rate reference limits are listed in Table VII. The range *rate* analog voltage reading should be averaged. Repeat for the remaining seven runs of Table VII. Note: For the last four inns, set the FUNCTION SELECT switch to the RANGE RATE REF (-) position.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

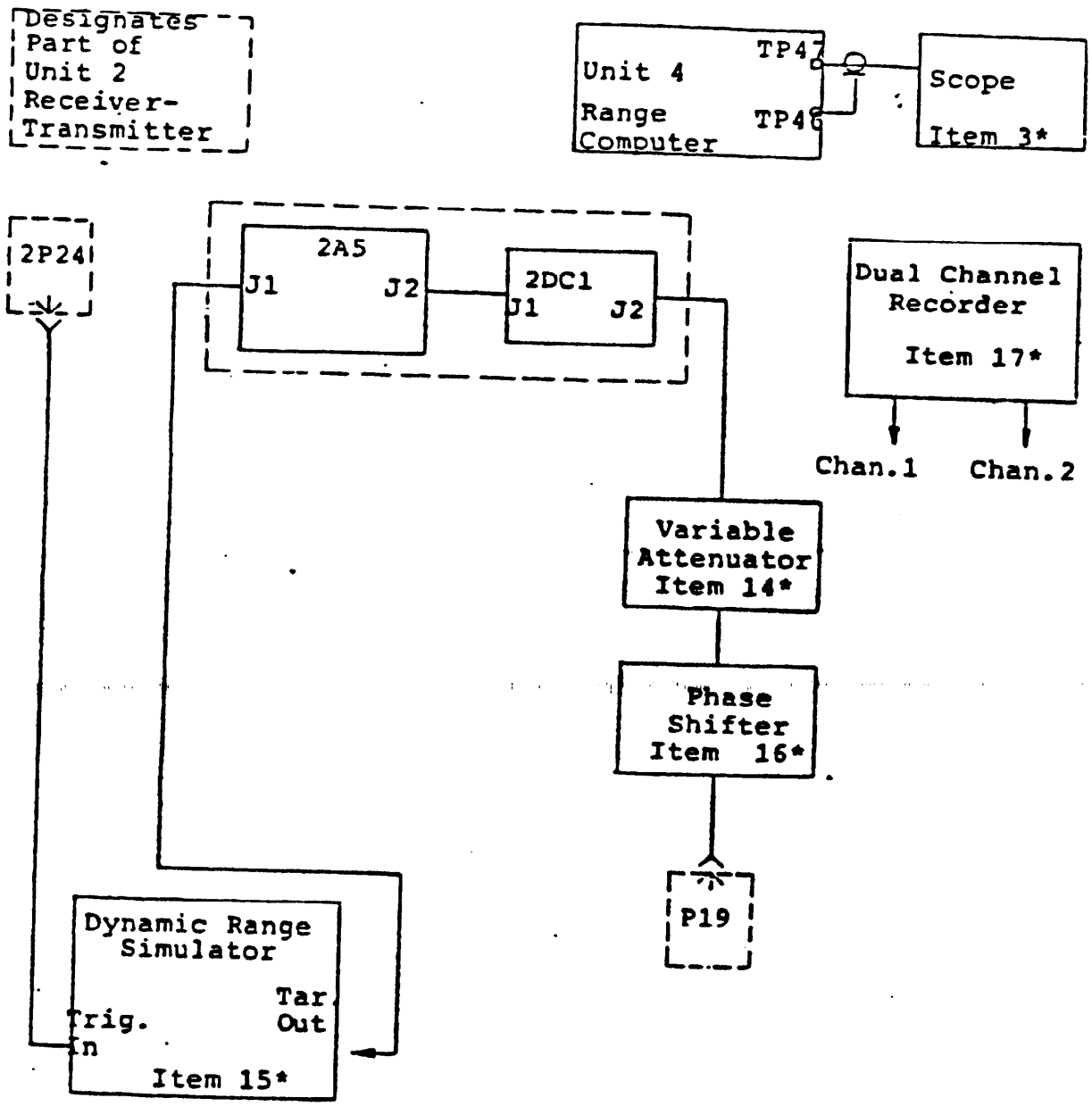


FIGURE 12. Range and range rate test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

TABLE VII. Range rate accuracy tests

RUN	START (m)	RANGE	STOP (ET)	VELOCITY FT/S	RANGE RATE REF. (v)	RATIO NLT	RATIO NMT
1	1,100	Increase	11,000	00918.7	+10	0.888	0.921
2	1,100	Increase	11,000	00697.2	+10	0.660	0.693
3	1,100	Increase	11,000	00344.2	+10	0.323	0.355
4	1,100	Increase	11,000	00057.4	+10	0.0404	0.0727
5	11,000	Decrease	1,000	00057.4	-lo	0.0404	0.0727
6	11,000	Decrease	1,000	00344.2	-lo	0.323	0.355
7	11,000	Decrease	1,000	00697.2	-lo	0.660	0.693
8	11,000	Decrease	1,000	00918.7	-lo	0.888	0.921

4.7.3.9.15 Three-second fade logic. with the test equipment set up the same as in 4.7.3.9.14 and with the radar in the standby mode, program the simulator as follows:

START	(FT)	1,000
STOP	(FT)	11,000
VELOCITY		INC. 100 FT/S
RANGE		INCREASING

Depress the RESET butto of the simulator. Connect the LEFT channel of the recorder to TP3 and TP5 (return) on the 4A21 PC card in unit 4 and right channel to the RANGE ANALOG SIGNAL and R-A return. With the deflection sensitivity of both channels set to 5 V/cm and the speed to 20 mm/s, turn on the recorder and press the START button on the simulator. On unit 4, press RATE switch S6 to RESET and hold and simultaneously press the FADE button on the simulator; hold both buttons for 5 seconds, then release both. The elapsed time between where RIGHT channel deflects and LEFT channel deflects must be 3.00 ± 0.25

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

4.7.3.9.16 one-second fade logic. The test equipment setup is the same as in 4.7.3.9.15. Use the precision dc power supply (Table IV, item 20) to calibrate the recorder charnels. Connect the precision supply to the LEFT channel. Set the recorder inputs as follows: Set the supply for +3.52 V to give a deflection sensitivity of 109 m/s per 10 mm. Disconnect the supply and connect a cable from RANGE RATE and R-R RETURN to the LEFT channel. Connect the precision supply to the RIGHT channel. Set the supply for +4.oo V (1000m) to give a deflection sensitivity of 1000 m/10 mm. Disconnect the supply and connect the cable from RANGE ANALOG SIGNAL and R-A RETURN to the RIGHT charnel. The recorder speed shall be 20 mm/s during the runs. With reference to Table VIII, program the simulator as specified in RUN 1. Press the RESET button on the simulator, then the START button. Start the recorder, then hold the FADE button on the simulator. The length of time that the radar remains in coast must be 1.0 ± 0.02 s.The range rate jump during the fade must be within ± 10 m/s. Repeat for remaining five runs of Table VIII .

TABLE VIII. one second fade logic tests

RUN	START FT	RANGE	STOP FT	VELOCITY FT/S
1	1,100	Increase	11,000	00328.1
2	1,100	InCrease	11 0Uoo	00656.2
3	1,100	Increase	11,000	00984.3
4	11,000	Decrease	1,100	00984.3
5	11,000	Decrease	1,100	00656.2
6	11,000	Decrease	1,100	00328.1

4.7.3.9.17 Range rate lag. The test setup is the same as 4.7.3.9.16. Set the dynamic range simulator as follows:

START	(FT)	12,000
sTOP	(FT)	10,782
ACCELERATION		270 FT/S^2
VELOCITY		o Ft/s
RANGE		DECREASE

Adjust the attenuator (Table IV, item 14), for 600 mV peak-to-peak; adjust the phase shifter (Table IV, item 16), for maximum signal; readjust the attenuator for 600 mv peak-to-peak. Turn on the recorder, allow the target to be acquired by the radar, then depress the START button on the simulator. At the end of the run the range rate must be a minimum of 217 m/s. Set the AN/VPM-2 POWER switch to OFF. Disconnect the AN/VPM-2 from the radar.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the current version before use.

4.7.3.9.18 *Minimum* range tracking. Set up test equipment per Figures 3, 5, and 10. Place radar in standby mode. Connect digital VM (Table IV, item 5) to RANGE ANALOG (4J2-J) and R-A RETURN (4J2-K) of the radar (TP7 and TP8 of the Range Information Monitor (Table IV, Item 27). Set target simulator (Table D/, item 15) for O FT/S velocity, START FT at 000999 FT, and zero acceleration. Press the reset button on the simulator. Decrease the range setting until the digital VM reads +1.04 Vdc *or less.* The PHASE SHIFTER (Table IV, item 16) may be adjusted if necessary to achieve the reading.

4.7.3.9.19 Receiver bandwidth. Set up the test equipment as shown in Figure 3. The radar is to be in the standby mode. with the rf signal generator (Table IV, item 22) set for 61.0 MHz, the output attenuator set for -3o dBm, and the output terminated in 50 ohms , connect the output to 3J1-D of the radar. With a 50-ohm termination on 3A1J2 (i-f amplifier assembly 10548164) of the radar and the rf voltmeter (Table IV, item 23) terminated in 50 ohms connected to 3AIJ3, tune the rf signal generator for maximum indication on the voltmeter. Adjusting the frequency of the generator higher than 61 MHz, note the frequency at which the voltmeter reads 3 dB lower than the maximum just obtained. Adjust the frequency of the generator below 61 MHz and note the frequency at which the voltmeter reads 3 dB lower than maximum. The difference between the two frequencies shall be 1.9 ± 0.2 MHz. On system control unit (Table IV, item 26) turn off PRIMARY POWER switch Sl.

4.8 Soldering. Soldering processes *shall* be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 GENERAL. Preservation, packing. and marking shall be in accordance Packaging Data Sheet PSI0549058. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. to be used to acquire and track moving aircraft targets and supply range and range rate information-to the fire control system for gun -lead angle computation.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:31Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of Conformance for each lot or shipment of products

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, & Army AMCCOM, ATTN: AMSMC-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (JWWDCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these *activities is now* under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

20 mm Anti-aircraft Artillery Gun
Fire Control System (FCS)
Radar Set AN/VPS-2
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. Asterisks (or *vertical* lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A733)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-50651B (AR)	26 December 1990

3. DOCUMENT TITLE

RADAR SET AN/VPS-2: 10549058 (RANGE ONLY RADAR)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

[redacted]

6. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:31Z
Check the source to verify that this is the current version before use.